

Draft: format for live data processing

Introduction

The format does not require any **headers** or other grouped forms (e.g., the Ethernet packet from TPX3 raw format). The format exposes both **pixel coordinates and extended timestamps** directly, with no headers required.

Additionally, it is important to have **common data** available in the format:

- All **timestamps** use the same clock
- Time is extended to the **same 56-bit range** in all packets
- All timestamps occupy the same bit positions across packet types, making time slicing trivial.

The specification utilizes **96-bit "words"** that are fully self-contained and do not require any global context to be unpacked.

The format

i Reporting Ranges: When reporting the ranges for each field, the bitfield definitions include both start and end. For example: 95 ... 65 is equivalent to [65, 95] in the [mathematical notation for intervals](#), indicating that it starts (and includes) 65 and goes up to and including 95.

Common part

1 | 95 ... 92 | 91 ... 36 | 35 ... 0 |

2 | Packet type | Timestamp | Packet specific data |

Field	Values	Size (bits)
Packet type	0 = Pixel 1 = TDC 2 = Control Other: future expansion	4

Timestamp 3840 MHz clock ticks 1 unit = 260.41666 ps	Pixel: ToA, TDC,Control: Timestamp Extended to ~220 days	56
Packet specific data	Check below.	36

Pixel

1	35 ... 26	25 ... 16	15 ... 6	5 ... 0	
2	ToT	X	Y	Reserved	

Field	Values	Size (bits)
ToT 40 MHz clock 1 unit = 25ns	Pixel ToT, no correction applied	10
X Layout is taken into account, but no rotation is supported	X position	10
Y Layout is taken into account, but no rotation is supported	Y position	10
Reserved		6

TDC

1	35 .. 30	29	28 .. 0	
2	Channel	Edge	Reserved	

Field	Values	Size (bits)
Channel	1: TDC-1 channel 2: TDC-2 channel Other: future expansion	6
Edge	0: TDC rise 1: TDC fall	1

Reserved		29
----------	--	----

Control

1		35	..	32		31	..	0		
2		Subtype				Reserved				

Field	Values	Size (bits)
Subtype	0: Shutter open 1: Shutter close 2: Heartbeat 3: Timestamp	4
Reserved	Reserved for future expansion.	32